Dissertation / Project / Project Work Title:
AI-Driven Customer Churn Prediction and Sentiment Analysis

Course No. : DSECLZG628

Course Title: Dissertation / Project / Project Work

Dissertation / Project /Project Work Done by:

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Degree Program: M.Tech Data Science

Research Area: Artificial Intelligence, Machine Learning, Natural Language Processing, and Data Engineering

Dissertation / Project Work carried out at:

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1. Broad Area of Work

The project falls under the broad domain of Artificial Intelligence and Data Science with a focus on customer analytics in the telecommunications sector. The work integrates Machine Learning, Natural Language Processing, and data engineering techniques to solve real-world business problems.

The key areas involved in this project include:

- Machine Learning: To predict customer churn using structured data
- Natural Language Processing: To analyze customer sentiment from textual data such as complaints and feedback
- Data Engineering: To build scalable data pipelines for processing large volumes of telecom data
- Artificial Intelligence: To combine predictive and analytical capabilities for decision-making

2. Background

In the highly competitive telecommunications industry, customer retention is a critical challenge. Organizations face significant revenue loss due to customer churn. Traditional churn prediction models rely heavily on structured data such as usage patterns and billing information but often fail to incorporate customer sentiment, which plays a crucial role in understanding customer behavior.

With the increasing availability of unstructured data such as customer complaints, chat logs, and feedback, there is a growing need for intelligent systems that can analyze both structured and unstructured data.

This project aims to address this gap by integrating machine learning-based churn prediction with sentiment analysis using Natural Language Processing techniques. The system will provide deeper insights into customer behavior and enable proactive retention strategies.

3. Objectives

The objectives of my project are as follows:

- To study and understand customer churn patterns in the telecom domain
- To build a machine learning model for predicting customer churn
- To perform sentiment analysis on customer feedback using NLP techniques
- To integrate structured and unstructured data for improved prediction accuracy
- To evaluate the performance of the developed models using appropriate metrics
- To demonstrate the effectiveness of AI-driven solutions in customer retention

4. Scope of Work

The scope of this dissertation includes the design and implementation of a system that predicts customer churn and analyzes customer sentiment.

The work will involve:

- Collection and preprocessing of telecom customer data
- Development of churn prediction models using machine learning algorithms
- Implementation of sentiment analysis on customer textual data
- Integration of both models into a unified system
- Evaluation of system performance based on accuracy and insights generated

The project focuses on building a prototype system that can be extended for real-world deployment in telecom environments.

5. Plan of Work

Phases	Start Date-End Date	Work to be done
Dissertation Outline	4 May 2026 – 10 May 2026	Literature review, problem definition, and preparation of dissertation outline
Design & Development	11 May 2026 – 15 June 2026	Data collection, preprocessing, development of churn prediction model and sentiment analysis system
Testing	16 June 2026 – 10 July 2026	Model testing, validation, performance evaluation, and refinement

Dissertation Review	11 July 2026 – 23 July 2026	Documentation review, incorporating feedback, and final improvements
Submission	24 July 2026 – 2 August 2026	Final report preparation and submission

6. Literature References

[1] T. Verbeke, D. Martens, C. Mues, and B. Baesens, “Building Comprehensible Customer Churn Prediction Models,” Expert Systems with Applications, 2012.

[2] S. Huang, X. Liu, X. Peng, and Z. Niu, “Customer Churn Prediction in Telecom Using Machine Learning,” IEEE Access, 2020.

[3] B. Liu, “Sentiment Analysis and Opinion Mining,” Morgan & Claypool Publishers, 2012.

[4] J. Brownlee, “Machine Learning Mastery With Python,” Machine Learning Mastery, 2016.

[5] Research articles on NLP and sentiment analysis techniques from recent journals and conferences.

7. Particulars of the Supervisor and Examiner

	Supervisor	Additional Examiner
Name	Ramyasri Gomatham	Samarasimha Reddy Noookareddy
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8. Remarks of the Supervisor

The proposed project focuses on an important problem in the telecommunications domain, namely customer churn prediction combined with sentiment analysis. The integration of machine learning and natural language processing techniques makes the study relevant and aligned with current industry trends.

The project demonstrates a clear understanding of the problem domain and proposes a feasible methodology for implementation. The expected outcomes indicate potential for practical application and further research.

The student has shown a good understanding of the concepts and possesses the capability to successfully complete the project.

Information about the Supervisor:

BIRLA INSTITUTE OF TECHNOLOGY & SCIENCE, PILANI

WORK INTEGRATED LEARNING PROGRAMMES (WILP) DIVISION

SECOND SEMESTER OF ACADEMIC YEAR 2021-2022

(INSERT COURSE NUMBER HERE) : (INSERT COURSE TITLE HERE) OUTLINE

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